



**PAN-ATLANTIC  
UNIVERSITY**

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## School of Science and Technology

**COURSE CODE:** DTS304 (STREAM 1)

**COURSE TITLE:** Database Management I

**EXAMINATION:** Test (Written C. A.)

**INSTRUCTION:** Answer all questions

**TIME:** 45 Minutes

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**Read the Case Study carefully and use it to answer Question 1**

### **Case Study: The Ultimate Pet Shop**

#### **Background:**

At the Ultimate pet shop, the database operations are guided by the following **business rule**:

A dog should be housed in exactly one location, and a location may not house any single dog or at times house different kinds of dog.

A dog may meet many clients, but a client can meet at least one dog; or many dogs before adoption. A record of the day a client visits a dog (the meeting date) is kept whenever the visitation happens.

Also, a client can adopt no dog or as many dogs as s/he wants. Conversely, a dog can only be adopted by a single client. For this relationship, the date of adoption should be recorded/stored in the database.

There are some additional information (the unique identifier in bold font):

1. A dog has the following attributes: **id**, name, arrivaldate, breed, dob, and temperament.
2. A location has the following attributes: **id**, name, addressLine, and capacity.
3. A client is identified by its **id**, name, phonenumber, and an email address.

#### **Question 1**

Draw the detailed entity relationship diagram (ERD) for the ultimate pet shop business entity. Pay attention to the following: **[10 marks]**

- i. The attributes of the entities must be shown and the unique identifier for each entity must be depicted.
- ii. Show the minimum and maximum cardinality constraints for each entity participating in a relationship.

#### **Question 2**

A local public library is struggling with its data management system. They currently use individual computer files to keep track of their books, library patrons, and lending records. For example, book information is in one file, patron details in another, and lending history in a third. Library staff find it cumbersome and difficult to get accurate, up-to-date information when they need to combine data from these different files, such as finding all books currently borrowed by a specific patron.

- (a) Identify and describe at least three distinct problems the library is likely facing with its current file-based system. **[4.5 marks]**
- (b) After solving the problem of the enterprise through the design of a DBMS and migrating their data into a proper database environment. Explain the activities of the query processing unit of your database engine. **[5.5 marks]**